

Topographic-Order Loss Headroom: A Mechanistic Demonstration

SegMAN-S · Sen1Floods11 · N = 100 · seed 0

Status: consolidated negative result, single-seed, label-relative, N=100 mechanistic. **Invariant respected throughout:** the DEM is **never** a model input. It appears only inside losses, diagnostics, and post-hoc analysis.

This document states the claim, the axioms, the definitions, one information-theoretic proposition, five empirical lemmas (each with the measured numbers), the composed argument, limitations, and — explicitly — what is *not* proven.

0. The precise claim (read this first)

Claim (bounded). For the pair (*SegMAN-S trained on Sen1Floods11 N=100 seed0, the reference labels y*), we evaluate the **shared mechanism** of the class of local monotone DEM-based topographic-order losses $\mathcal{C}_{\text{ord}}$ (see D5). In this setting the unconstrained Dice+CE baseline already satisfies the targeted local topographic-order property — as measured by violation fraction, violation energy, and distributional tails — **at least as strongly as the reference labels**. Local monotone DEM-based topographic-order losses whose sole mechanism is to reduce v_{ij} violations therefore have **no native label-relative headroom** in this model–data configuration.

What the claim is **not**:

- It is **not** "the DEM is useless for flood mapping in general." (It is not — see L4: a DEM-only predictor reaches AUC 0.77–0.84.)
 - It is **not** "no physics-informed loss can ever help any model." (Out of scope; see §10.)
 - It is **not** a statement about HAND, flow accumulation, hydrological connectivity, rainfall/discharge constraints, or DEM used as an input channel. All of these are explicitly outside scope (§10).
 - It is **not** "the model understands hydrology." (The opposite — see L3.)
 - It is **not** a statement about the *true physical flood state*; everything is **relative to the reference labels** (Axiom A3).
 - It is **not** a full-dataset or multi-architecture result. The N=100 setup is preliminary/mechanistic; §10 lists what remains before publishable claims.
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1. Notation and objects

symbol	meaning
$x \in \mathcal{X}$	model input: 15 channels (13 S2L1C + 2 S1GRD)
$y \in \{0, 1\}^{H \times W}$	reference water label
$z \in \mathbb{R}^{H \times W}$	DEM (elevation), never an input to f_θ
$f_\theta : \mathcal{X} \rightarrow [0, 1]^{H \times W}$	SegMAN-S; output $\hat{p} = f_\theta(x) = p_{\text{water}}$
$\hat{y} = \mathbb{1}[\hat{p} > \frac{1}{2}]$	binarised prediction
$\mathcal{L}_{\text{seg}}$	Dice + CE segmentation loss
$\mathcal{R}_\phi(\hat{p}, z)$	a DEM/physics regularizer of formulation ϕ (e.g. D8)
λ	physics-loss weight; total loss $\mathcal{L}_{\text{seg}} + \lambda \mathcal{R}_\phi$
splits	val (86 tiles), test (89), bolivia (15, OOD)

Checkpoint of record: segman_n100_dice_ce_seed0 , best epoch 34, val mIoU 0.8546.

2. Axioms (assumptions made explicit)

A1 — Channel of action. A DEM loss can influence the learned parameters *only* through the gradient $\nabla_\theta \mathcal{R}_\phi(f_\theta(x), z)$. It introduces no new input pathway (the DEM is not fed to f_θ).

A2 — Two and only two ways to help. Relative to a fixed label set, adding $\lambda \mathcal{R}_\phi$ can reduce held-out risk only via: - **(I) the information channel** — $\mathcal{R}_\phi$ injects label-relevant information that is carried by z but not already present in $f_\theta(x)$; or - **(II) the inductive-bias channel** — $\mathcal{R}_\phi$ shrinks the hypothesis space / acts as a prior / reduces variance, improving generalization *even with zero new information*.

Any real effect decomposes into these two. (A1–A2 are the logical backbone; everything below bounds channel I to ≈ 0 and tests channel II for the class $\mathcal{C}_{\text{ord}}$.)

A3 — Label-relative evaluation. All risks, AUCs and mutual-information quantities are defined against the reference labels y , **not** the unobserved true flood state y^* .

A4 — Two regimes. *Post-hoc* diagnostics (L3, L4) hold θ **frozen**. The *loss* experiment (L1) **retrains** θ . This distinction matters: post-hoc tests bound what z adds *given the converged model*; the retraining test bounds what z adds *when injected as a constraint during training*.

A5 — Finite-sample honesty. Every empirical quantity is an estimate. We report bootstrap CIs (L1b, L4) or per-tile dispersion (L2) and never treat a point estimate as an exact zero.

3. Definitions

D1 — Output-level constraint headroom. Let $A(s; w)$ be the AUROC of a real-valued score s for a binary target w . Using **negative elevation** $-z$ as the score (water sits low $\Rightarrow -z$ predicts water),

$$H_z := A(-z; y) - A(-z; \hat{y}).$$

$H_z > 0$: the *labels* are more topographically separable than the *predictions* — room for a topographic loss to push. $H_z \leq 0$: the predictions are already at least as topographically aligned as the labels — **no room**.

D2 — Representational decodability. Let $\phi(x)$ be frozen features and $D(\phi)$ the best *within-tile linear* decodability (Pearson r , per-tile z-scored) of z from ϕ . Compare ϕ_{trained} , ϕ_{random} (random-init same arch), ϕ_{input} (raw 15-ch). If $D(\phi_{\text{trained}}) \approx D(\phi_{\text{input}})$, training induced **no privileged** elevation representation.

D3 — Conditional DEM redundancy. The DEM is *conditionally redundant given the model output* iff

$$I(y; z | \hat{p}) = 0.$$

Estimator: the held-out AUC gap $\Delta\text{AUC} = \text{AUC}(y \sim \hat{p} + z) - \text{AUC}(y \sim \hat{p})$ of a flexible learner (gradient boosting). $\Delta\text{AUC} \approx 0$ (CI spanning 0) $\Rightarrow I(y; z | \hat{p}) \approx 0$.

D4 — Effective constraint contribution. $\rho := \lambda \mathcal{R}_\phi / (\mathcal{L}_{\text{seg}} + \lambda \mathcal{R}_\phi)$ at convergence — the share of the training objective actually supplied by the physics term. A *slack* constraint (already satisfied) has $\rho \approx 0$ and exerts almost no gradient.

D5 — Local topographic-order loss class $\mathcal{C}_{\text{ord}}$. We do **not** refute D4 and D8 independently. We evaluate the **shared mechanism** of their class: reducing local topographic-order violations.

$$\mathcal{C}_{\text{ord}} = \{ \mathcal{R}_\phi : \mathcal{R}_\phi \text{ penalises } p_i > p_j + \tau \text{ when } h_i > h_j, j \in \mathcal{N}(i) \}$$

where h_i is DEM elevation, j is a lower neighbouring pixel, and τ is a tolerance margin. The class includes: D4 (all lower 4-neighbors), D8 (steepest-descent downstream), slope-weighted D8, hinge-linear/quadratic variants, margin-based variants, and any local loss whose main mechanism is reducing these violations. **Not included:** DEM-as-input, HAND, flow accumulation, hydrological connectivity, or any non-local DEM-based constraint.

D6 — Local violation magnitude and violation fraction. For a valid edge (i, j) with $h_i > h_j$:

$$v_{ij}(\hat{p}, z) = \max(0, p_i - p_j - \tau)$$

The (slope-weighted) **violation fraction** (VF) over the D8 active set $\mathcal{E}$:

$$\text{VF}(\hat{p}, z) = \frac{\sum_{(i,j) \in \mathcal{E}} w_{ij} \mathbb{1}[p_i > p_j + \tau]}{\sum_{(i,j) \in \mathcal{E}} w_{ij}}$$

with $w_{ij} = \min(1, \text{drop}_{ij}/s_0)$ (D8 parameters: $s_0 = 1.0$, $\tau = 0.05$). The **native headroom** is $H_R = \text{VF}(\hat{y}_0, z) - \text{VF}(y, z)$. $H_R \leq 0$ means the baseline violates the constraint no more than the labels.

D7 — Violation energy E_{topo} .

$$E_{\text{topo}}(p, z) = \frac{\sum_{(i,j) \in \mathcal{E}} w_{ij} [\max(0, p_i - p_j - \tau)]^2}{\sum_{(i,j) \in \mathcal{E}} w_{ij}}$$

This is the weighted mean squared violation — matching the D8 loss objective exactly.

4. Central proposition — the information bound

Proposition 1. Let ℓ be any proper scoring rule (e.g. log-loss or Brier), whose Bayes predictor is the conditional mean. If $I(y; z | \hat{p}) = 0$ — equivalently $y \perp z | \hat{p}$ — then for the Bayes risks

$$R^*(\hat{p}, z) = R^*(\hat{p}),$$

i.e. **no measurable function of $(\hat{p}, z)$ predicts y better than the best function of $\hat{p}$ alone.**

Proof. Conditional independence gives $\mathbb{E}[y | \hat{p}, z] = \mathbb{E}[y | \hat{p}]$ almost surely. For a proper loss the risk-minimizing predictor is the conditional expectation, so the optimal predictors — and hence their risks — coincide. $\square$

Corollary 1 (information channel). Under A1–A2, a DEM loss acting through channel I can lower held-out risk only to the extent that $I(y; z | \hat{p}) > 0$. If that conditional information is zero, the information channel has **exactly zero** label-exploitable headroom, *for every formulation ϕ* — because all formulations act through the same z and can supply nothing z does not contain.

Scope of Proposition 1. It conditions on a *fixed* $\hat{p}$. It therefore bounds the **post-hoc / converged-model** information channel. The path it does *not* close by itself is "the constraint, applied *during* training, reshapes $\hat{p}$ into something better." That residual path is exactly what Lemma L1 (actual D8 retraining) and Lemma L1b (class-level mechanistic diagnostics) test.

5. Empirical lemmas

Lemma L1 — the direct loss result (channel II, D8 formulation)

D8 downstream topographic loss, seed 0, $\lambda \in \{1, 100\}$, real vs shuffled DEM control. mIoU (↑ better); precision/recall on water; predicted-mask topographic-violation fraction (VF).

split	metric	baseline	D8 real $\lambda 1$	D8 shuf $\lambda 1$	D8 real $\lambda 100$
val	mIoU	0.8546	0.8586 (+0.004)	0.8401	0.8398 (−0.015)
test	mIoU	0.8615	0.8600 (−0.001)	0.8460	0.8579 (−0.004)
bolivia	mIoU	0.8434	0.8419 (−0.001)	0.8379	0.8408 (−0.003)
test	precision	0.8680	0.9026 (+0.035)	0.8927	0.8620
test	recall	0.8604	0.8243 (−0.036)	0.8036	0.8586
test	topo-VF	0.00110	0.00091	0.00079	0.00094

Two decisive facts:

- The constraint is slack.** Effective contribution $\rho = 7.1 \times 10^{-5}$ at $\lambda = 1$ and 3.0×10^{-3} at $\lambda = 100$ — i.e. **0.007 % / 0.3 %** of the objective. The baseline already violates the topographic constraint on only $\approx 0.1\%$ of pixels (VF ≈ 0.0011). There is almost nothing for the penalty to act on.
- No robust held-out/OOD gain.** D8 real improves `val` by +0.004 mIoU but *costs* −0.001 on both `test` and `OOD bolivia`. Mechanistically it merely trades **recall for precision** (test: +0.035 precision, −0.036 recall) — it shrinks the water mask, not improves it. $\lambda = 100$ makes every split worse.

Real vs shuffled is not a robust physics signal. At $N=100$ $\lambda 1$ real edges out shuffled, but the shuffled run converged earlier (best epoch 22 vs 31) and an earlier $N=50$ topographic-loss study found the **opposite** sign (shuffled beat real on all splits). Across settings the sign of (real − shuffled) **flips** and is single-seed. Combined with $\rho \approx 0$, the topographic *content* provides no reproducible benefit beyond generic regularization. ■(L1)

Lemma L1b — class-level mechanistic headroom: $\mathcal{C}_{\text{ord}}$ (channel II)

L1 tested one formulation (D8, seed0). L1b evaluates the **shared mechanism** of the entire class $\mathcal{C}_{\text{ord}}$ (D5): comparing baseline predictions and reference labels in the native violation space of the constraint. All measurements are inference-only, checkpoint frozen.

5.1 D8 native violation-fraction headroom

$$H_R = \text{VF}(\hat{y}_0, z) - \text{VF}(y, z), \text{ slope-weighted, } s_0 = 1.0, \tau = 0.05.$$

split	VF(labels)	VF(pred)	H_R (pooled)	per-tile mean	95% CI (per-tile)	% tiles pred $\leq$ labels
val	0.00217	0.00062	-0.00154	-0.00228	[-0.00303, -0.00159]	88%
test	0.00241	0.00058	-0.00182	-0.00268	[-0.00344, -0.00120]	90%
bolivia	0.00321	0.00094	-0.00227	-0.00247	[-0.00406, -0.00105]	80%

Note on the two estimators. The *pooled* H_R weights tiles by their activity count (more active tiles contribute more); the *per-tile mean* and its bootstrap CI weight each tile equally. For val, the pooled value (-0.00154) lies just above the upper bound of the per-tile CI (-0.00159) because large/active tiles tend to have smaller magnitude H_R , pulling the pooled estimator toward zero. **Both estimators are negative; all per-tile CIs are entirely negative.** The conclusion is unchanged: the baseline violates D8 3–4× less than the labels.

Interpretation: the negative H_R indicates no native VF headroom for any D8-style loss. The predictions are already more locally topographic-order consistent than the reference masks.

5.2 D4 native headroom

D4 uses all lower 4-neighbours (not just steepest descent). It belongs to $\mathcal{C}_{\text{ord}}$ by the same mechanism. Results (same parameters):

split	VF _{D4} (labels)	VF _{D4} (pred)	H_R^{D4}	% tiles pred $\leq$ labels
val	0.00180	0.00055	-0.00125	88%
test	0.00204	0.00051	-0.00153	90%
bolivia	0.00284	0.00082	-0.00202	80%

The no-headroom result is not an artifact of the steepest-descent neighbor selection. It holds under the simpler 4-neighbor local formulation. $H_R^{\text{D4}} \in [-0.00202, -0.00125]$ across splits. ■(D4 sub-result)

5.3 Violation energy E_{topo}

split	E_{topo} (labels)	E_{topo} (pred $\hat{y}$)	ΔE	E_{topo} (pred soft p)
val	0.001955	0.000564	-0.001391	—
test	0.002174	0.000527	-0.001647	—
bolivia	0.002896	0.000849	-0.002047	—

The baseline has approximately **3× lower violation energy** than the labels. The absence of headroom is not only visible in a binary violation count; it holds in the squared-severity violation energy that the D8 loss directly minimises. ■(E_{topo} sub-result)

5.4 Distributional dominance: $P(v > t)$

A critic could argue that lower mean VF is insufficient — the baseline might still have severe violations that a weighted loss could target. We compare empirical tail probabilities $P(v_{\hat{p}_0} > t)$ vs $P(v_y > t)$ for $t \in \{0, 0.05, 0.10, 0.20, 0.30, 0.50, 0.70\}$:

$$P(v_{\hat{p}_0} > t) \leq P(v_y > t)$$

holds at **all tested thresholds on all three splits** (val, test, bolivia/OOD). This is stronger than a mean comparison: the baseline violates the local topographic-order constraint less than the labels across the full

range of tested violation severities, weakening the objection that stronger severity weighting could recover headroom. ■ (distributional dominance)

5.5 Useful violation rate: real signal, no net benefit

Among active D8 violations in the baseline prediction — the set the D8 loss would act on — we classify the centre pixel i by the ground truth:

- **Useful** ($\hat{y}_i = 1, y_i = 0$): suppressing p_i removes a false positive.
- **Harmful** ($\hat{y}_i = 1, y_i = 1$): suppressing p_i removes true water (recall loss).
- **GT-endorsed**: the GT mask itself has $y_i = 1, y_{d(i)} = 0$ at this location — the loss penalises a label-correct configuration.

split	n violations	useful	harmful	GT-endorsed	enrichment
val	23,168	58.6%	41.4%	8.9%	~3.9×
test	21,950	58.6%	41.4%	10.2%	~4.4×
bolivia	5,875	60.9%	39.1%	6.9%	~3.7×

Enrichment = $P(\text{GT dry} \mid \text{D8-active violation}) / P(\text{GT dry} \mid \text{predicted water}) \approx 3.7\text{--}4.4\times$.

Interpretation. The D8 signal is real but not net-useful in this label-relative setting. Topographic violations are enriched in false positives (~4× above baseline FP rate). However, approximately 41% of corrections would suppress true water. This precisely explains the observed pattern from L1: enforcing D8 raises **precision** (+0.035) but costs **recall** (−0.036), leaving IoU essentially unchanged. ■ (L1b)

Lemma L2 — output-level headroom is non-positive (channel I, output)

Pooled $A(-z; \cdot)$ from `elevation_auc_predictions_segman_n100_dice_ce_seed0`:

split	$A(-z; y)$	$A(-z; \hat{y})$	H_z	per-tile std
val	0.7537	0.7453	+0.008	≈ 0.18
test	0.7615	0.7684	−0.007	≈ 0.18
bolivia	0.5664	0.5929	−0.027	≈ 0.10

Sanity controls in the same run: shuffled-elevation AUC = **0.500** (no leakage); model-vs-GT AUC = **0.985** (the segmentation itself is excellent). $H_z \leq 0$ on `test` and OOD `bolivia`; the lone positive value (`val`, +0.008) is two orders of magnitude below the per-tile std (0.18) — i.e. indistinguishable from zero. **The predictions are already as topographically aligned as the labels.** ■ (L2)

Lemma L3 — no privileged elevation representation (channel I, representation)

Within-tile linear DEM-decodability probe, frozen backbone, Pearson r (pooled):

split	trained	random	input
val	0.414	0.430	0.405
test	0.368	0.348	0.372
bolivia	0.337	0.367	0.474

$D(\phi_{\text{trained}}) \approx D(\phi_{\text{random}}) \approx D(\phi_{\text{input}})$ on every split; on OOD `bolivia` the *raw input* decodes elevation best. A random-init backbone matches the trained one. **Segmentation training induced no privileged elevation representation** for a constraint to strengthen; the weak ($r \approx 0.4$) decodability is passive input-level optical/SAR $\leftrightarrow$ terrain correlation that any network passes through. *Caveat*: linear probe; but `trained  $\approx$  random  $\approx$  input` collapses the claim to "training adds nothing over the input." ■(L3)

Lemma L4 — conditional redundancy: the formulation-independent leg (channel I, all formulations)

Gradient boosting predicts y ; $A : y \sim \hat{p}$, $B : y \sim \hat{p} + z_{\text{DEM}}$, $C : y \sim z_{\text{DEM}}$, $D : y \sim \hat{p} + \text{shuffle}(z)$. Bootstrap-by-tile 95% CI.

split	AUC(A)	AUC(B)	$\Delta\text{AUC} = B - A$	95% CI	DEM-only (C)
val	0.9840	0.9861	+0.0021	[+0.0008, +0.0043]	0.8366
test	0.9848	0.9840	-0.0009	[-0.0038, +0.0016]	0.8336
bolivia	0.9817	0.9822	+0.0004	[-0.0014, +0.0029]	0.7660

- $\Delta\text{AUC} \approx 0$ everywhere while **DEM-only AUC = 0.77–0.84** $\Rightarrow$ genuine *conditional* redundancy, not "DEM is uninformative." This is the empirical estimate of $I(y; z \mid \hat{p}) \approx 0$.
- **Sanity** D (shuffled DEM) = A exactly. **Feature importance** (model B): $\hat{p} = 0.39$, every DEM feature ≤ 0.002 .
- **Anti-ceiling** (uncertain pixels $0.05 < \hat{p} < 0.95$, where there is room): $\Delta\text{AUC} = +0.011$ (val) / **-0.021 (test)** / +0.004 (bolivia) — signs flip, DEM even *hurts* on test. The null is not a ceiling artifact.

Because gradient boosting can model arbitrarily complex $z \leftrightarrow y$ interactions, it upper-bounds *any* DEM-loss formulation's information channel. It finds ≈ 0 . ■(L4)

6. The composed argument

Information channel (I). By Proposition 1, channel I is governed by $I(y; z \mid \hat{p})$. - L2 bounds it to ≤ 0 at the **output** level (headroom non-positive). - L3 bounds it to ≈ 0 at the **representation** level (no privileged encoding). - L4 estimates it directly as ≈ 0 for **all formulations** (flexible learner, with CIs, anti-ceiling, and a working negative control).

These three are independent measurements at three levels (prediction, features, post-hoc information) and they agree. $\Rightarrow$ **Channel I has no label-exploitable headroom in this model–data pair.**

Inductive-bias channel (II), class $\mathcal{C}_{\text{ord}}$. L1 and L1b together close channel II for this class in this setting:

- L1: D8 direct training shows $\rho \approx 0$, no robust test/OOD gain, precision $\leftrightarrow$ recall trade only.
- L1b §5.1: $H_R < 0$ on all splits (per-tile CI entirely negative) — baseline already violates D8 3–4 $\times$ less than labels.
- L1b §5.2: $H_R^{\text{D4}} < 0$ — same result under 4-neighbor formulation.
- L1b §5.3: $E_{\text{topo}}(\hat{y}_0) < E_{\text{topo}}(y)$ by $\approx 3\times$ — holds in the severity-weighted energy.
- L1b §5.4: distributional dominance $P(v_{\hat{p}_0} > t) \leq P(v_y > t)$ at all thresholds.
- L1b §5.5: 41% of active corrections suppress true water, explaining the observed precision/recall trade without net IoU benefit.

$\Rightarrow$ **For the class $\mathcal{C}_{\text{ord}}$, channel II provides no net label-relative benefit in this setup. The signal is real ($\sim 4\times$ FP enrichment) but not net-useful (41% collateral).**

Theorem (bounded, label-relative, class-level). For (*SegMAN-S*, $N=100$ seed0, reference labels y): (i) no DEM loss of any formulation has label-exploitable headroom through the information channel (Prop 1 + L2 + L3 + L4); and (ii) the class $\mathcal{C}_{\text{ord}}$ of local monotone topographic-order losses exhibits no net native label-relative headroom (L1 + L1b). **On this task, a local monotone DEM-based topographic-order loss is not justified.**

The null is thus an **informational and mechanistic property of the model–data pair**, not a failure to find the right formula within this class.

7. What is NOT proven (the open doors — state these explicitly)

- Channel II for *other* formulations / loss classes.** L1 + L1b close channel II for $\mathcal{C}_{\text{ord}}$ (D4, D8, VF-monotone variants). Losses based on a **different mechanism** — hydrological connectivity, flow accumulation, HAND, non-local constraints — are outside $\mathcal{C}_{\text{ord}}$ and are **not tested**. The theorem does not apply to them.
- True physical state vs labels (A3).** Every lemma is label-relative. If the labels are topographically noisy, a DEM loss could still improve agreement with the unobserved truth y^* . None of L1–L4 can see this; the 41% GT-endorsed violations in §5.5 hint that some "corrections" may in fact be toward y^* .
- DEM as a free input modality.** All evidence concerns the DEM as a *constraint/feature*. The strongest test of channel I — retraining with the DEM as an input channel, letting it reshape the representation end-to-end — is **not yet run**. L4 conditions on a *frozen* $\hat{p}$ and so cannot exclude that a model trained *with* DEM would have built a different $\hat{p}$.
- Single seed / missing positive control.** All training runs are seed 0. The definitive positive control (an S1-only or early/undertrained checkpoint where ΔAUC *should* turn clearly positive) does not exist.

8. Falsifiability (what would overturn this)

observation	which lemma it breaks	consequence
A loss in $\mathcal{C}_{\text{ord}}$ gives reproducible multi-seed test/OOD gain	L1 / Theorem(ii)	channel II is live for that formulation
$H_R > 0$ or per-tile CI not entirely negative	L1b §5.1	VF headroom exists
$E_{\text{topo}}(\hat{y}_0) > E_{\text{topo}}(y)$	L1b §5.3	energy headroom exists
$P(v_{\hat{p}_0} > t) > P(v_y > t)$ for some threshold	L1b §5.4	tail headroom exists
$\Delta\text{AUC}(B-A)$ CI clearly > 0 on a held-out split	L4 / Prop-1 premise	conditional information exists
H_z robustly > 0 beyond per-tile std on test/OOD	L2	output headroom exists
DEM-as-input beats S1/S2 (and shuffled), multi-seed	open door #3	the modality carries exploitable signal
Cleaner labels reveal a DEM gain vs y^*	open door #2 (A3)	claim is label-artifact

A single such result falsifies the corresponding leg. As of this writing none is observed.

9. Conclusion

Precise defensible claim:

"We do not refute D4 and D8 independently. We evaluate the shared mechanism of their loss class: reducing local topographic-order violations. In the N=100 SegMAN-S/Sen1Floods11 setting, the unconstrained Dice+CE baseline already exhibits lower violation fraction, lower violation energy, and lower violation tails than the reference labels. Active D8 violations are enriched in false positives ($\sim 4\times$ above baseline FP rate), showing that the signal is real, but approximately 41% of corrections would suppress true water, explaining the observed precision–recall trade-off and lack of net IoU improvement. Therefore, local monotone DEM-based topographic-order losses whose sole mechanism is to reduce D4/D8/VF violations have no native label-relative headroom in this setup. This is an N=100, seed-0, label-relative result; §10 lists what remains before a publishable claim."

Sentences to avoid (a reviewer will break them): - "The DEM is useless for flood segmentation." (L4-C: DEM-only AUC 0.77–0.84.) - "No physics-informed loss can ever help." (Channel II, other loss classes — open door #1.) - "The model learned/understands elevation." (L3: the opposite.) - "DEM input useless $\Rightarrow$ DEM loss useless." (Skips channel II; input test not yet run.) - "No DEM-based method can ever work." (Scope is $\mathcal{C}_{\text{ord}}$ only.)

Operational consequence. Stop iterating formulations within $\mathcal{C}_{\text{ord}}$ on this SegMAN N=100 pair. Remaining high-value moves: (a) DEM-as-input ablation + S1-only positive control to close open doors #3 and #4; (b) porting the headroom/redundancy screen to EoMT to show it generalises as an a-priori test, turning this negative result into a method.

10. Limitations

- N=100 is preliminary/mechanistic.** These experiments demonstrate the mechanism on a controlled low-data setup. Publishable claims require full Sen1Floods11 experiments with multi-seed confirmation across architectures.
- Single seed for training experiments.** L1 (direct loss result) is seed 0 only. Multi-seed confirmation (target: 5 seeds $\times$ 3 conditions, TOST equivalence testing with $\delta \approx \sigma_{\text{seed}} \approx 0.003\text{--}0.005$ water-IoU) is a required next step. L1b diagnostics are inference-only and do not depend on training seed.
- Label-relative (Axiom A3).** All results are relative to the available reference labels. If labels are systematically noisy in topographically complex areas, the DEM could improve agreement with the true physical state $\mathbf{y}^*$ without improving agreement with $\mathbf{y}$. The $\sim 8\text{--}10\%$ GT-endorsed violations (§5.5) suggest some label noise is present.
- Scope: local monotone topographic-order losses only.** The conclusion applies to $\mathcal{C}_{\text{ord}}$ — losses penalising $p_i > p_j + \tau$ when $h_i > h_j$ for neighbouring pixels. It does **not** apply to: DEM as input · HAND · flow accumulation · hydrological connectivity losses · rainfall/discharge constraints · temporal hydrodynamic constraints · non-local or orthogonal DEM-based methods.
- DEM as input not tested.** This study concerns DEM as a constraint only. A model trained with DEM as an input channel might build a different $\hat{p}$, for which L4 would need to be re-evaluated.
- Spatial resolution and annotation limits.** The Copernicus GLO-30 DEM is a DSM (30 m resolution, includes buildings/canopy, not bare earth). D8/D4 local rules at 30 m may flag legitimate flood configurations as violations — backwater, levee-contained water, urban drainage, rainfall-driven ponding, flat floodplain ambiguity — contributing to the 41% harmful and 7–10% GT-endorsed violation rates.

7. **Single architecture.** Results are for SegMAN-S. Whether the mechanistic screen generalises to other architectures (EoMT, ViT-based models) is an explicit next step.

11. Environment and reproducibility

We use the full SegMAN-S architecture adapted to 15-channel Sentinel-1/Sentinel-2 inputs. In the Windows/PyTorch environment, optimised CUDA extensions are replaced by PyTorch fallback operators for compatibility and reproducibility. **The DEM is never used as a model input.**

Check	Result
OS	Windows 10 Pro for Workstations, build 26200
Python	3.11.9 (MSC v.1938 64-bit)
PyTorch / CUDA	2.5.1+cu121 / CUDA 12.1
GPU	NVIDIA RTX 5000 Ada Generation, 32 GB
SegMAN-S parameters	33,447,272 (33.45M)
Input shape	(1, 15, 512, 512)
Output shape	(1, 2, 512, 512)
Official CUDA extensions	not installed (mamba-ssm, natten, mmcv-full)
PyTorch shims	active: selective scan (exact), cross-scan (exact), NATTEN (exact on interior pixels)
DEM as model input	never
N=100 checkpoints	41 runs, all with <code>best_checkpoint.pt</code> (~767 MB each)
Reproducibility report	<code>E:\flood_research\setup_logs\windows_segman_env\windows_segman_reproducibility_report.md</code>

NATTEN equivalence was verified for interior pixels; boundary behaviour (~1-pixel border ring) is documented in the shim source (`segman_kernels/compat.py`) and is negligible at 512×512.

Main project repo: commit `8017b883` on `https://github.com/KarimKQF/physics-informed-flood-segmentation.git`

SegMAN external source: commit `9ced66ab` (CVPR 2025 official)

12. Provenance (artifacts backing each number)

- **L1:** `reports/segman_n100_d8_seed0_results.json` , `reports/segman_n100_d8_lambda100_seed0_results.json`
- **L1b §5.1–5.2:** `outputs/native_vf_headroom/native_vf_headroom_results.json`
- **L1b §5.2–5.4 (D4/E_topo/tails):**
`outputs/topographic_order_headroom/topographic_order_headroom_results.json`
- **L2:** `reports/elevation_auc_predictions_segman_n100_dice_ce_seed0.json`

- **L3:** docs/dem_decodability_segman_n100_dice_ce_seed0.md ,
reports/dem_decodability_segman_n100_dice_ce_seed0.json
- **L4:** outputs/conditional_dem_redundancy/conditional_dem_redundancy_report.md (+ .json , .csv)
- **Earlier N=50 control (real-vs-shuffled sign flip):** reports/segman_seed0_loss_comparison_summary.json
- **Smoke test / env:** E:\flood_research\setup_logs\windows_segman_env\segman_smoke_test_output.txt